6. Factorisation of Algebraic expressions

① $x^2+9x+18$
$\Rightarrow = x^2+3x+6x+18$
$= \cancel{6x}\, x(x+3)+6(x+3)$
$= (x+3)(x+6)$

② $x^2-10x+9$
$= x^2-9x-1x+9$
$= x(9-x)+1(9+x)$
$= (9-x)(x+1)$

③ $y^2+24y+144$
$= y^2+12y+12y+144$
$= y(y+12)\,12+\cancel{y}(y+12)$
$= (y+12)(y+12)$

④ $5y^2+5y-10$
$= y^2+5y+10y-10$
$=$

⑤ $p^2-2p-35$
$p^2-7p-5p-35$
$p(7+p)\,5(7+p)$
$=(7+p)(p-5)$

⑥ $p^2-7p-44$
$= p^2-4p-11p-44$
$= p(4+p)+11(4+p)$
$= (4+p)(p-11)$

⑦ $m^2-23m+120$
$= m^2-15m+8m+120$
$= m(15-m)\,8+(15-m)$
$= (15-m)(m+8)$

⑧ $m^2-25m+100$
$= m^2-20m+5m+100$
$= m(20+m)+5(20+y)$
$= (20+m)(m+5)$

⑨ $2x^2+x-45$
$= 3x^2+9x+5x+15$
$= 3x(x+9)+5(x+3)$
$= (x+9)(3x+5)$

⑩ $3x^2+14x+15$
$= 3x^2+9x+5x+15$
$= 3x(3+x)\,5(3+x)$
$= (3+x)(3x+5)$

⑫ 44^2-x-3
$\rightarrow 44^2-11x-12x-3$
$=11(4-x)\,3(4-x)$
$= (4-x)(11-3)$

(11) $20x^2 - 26x + 8$

$= 20x^2 - 16x - 10x + 8$

$= 4x(5x-4)\ 2(5x+4)$

$= (5x-4)\ (4x-2)$

$\left\{ 20 \times 8 = 160 \right.$

Factors of a^3+b^3 :-

$$\boxed{a^3+b^3 = (a+b)\ (a^2-ab+b^2)}$$

Practice set 6.2

1. Factorise.

(1) $x^3 + 64y^3 = x^3 + (4y)^3$

$\rightarrow = (x+4y)\left[x^2 + x \times 4y + (4y)^2\right]$

$= (x+4y)\left(x^2 - 4xy + 16y^2\right)$

(5) $24a^3 + 81b^3 = 3(8a^3 + 27b^3)$

$\rightarrow = 3\left[(2a)^3 + (3b)^3\right]$

$= 3(2a+3b)\left[(2a)^2 - (2a \times 3b + (3b)^2\right]$

$= 3(2a+3b)\ (4a^2 - 6ab + 9b^2)$

(6) $y^3 + \frac{1}{8y^3} = y^3 + \frac{1}{(2y)^3}$

$= \left(y + \frac{1}{2y}\right)\left[y^2 - y \times \frac{1}{2y} + \left(\frac{1}{2y}\right)^2\right]$

$= \left(y + \frac{1}{2y}\right)\left(y^2 - \frac{1}{2} + \frac{1}{4y^2}\right)$

② $125p^3 + q^3 = (5p)^3 + q^3$

$= (5p+q)\left[(5p)^2 - 5p \times q + q^2\right]$

$= (5p+q)\left[25p^2 - 5pq + q^2\right]$

③ $125k^3 + 27m^3 = (5k)^3 + (3m)^3$

$\rightarrow$ $= (5k+3m)\left[(5k)^2 - 5k \times 3m + (3m)^2\right]$

$= (5k+3m)\left[25k^2 - 15km + 9m^2\right]$

④ $2l^3 + 432m^3 = 2(1l^3 + 216m^3)$

$\rightarrow$ $= 2(1l^3 + 6m^3)$

$= 2(1l+6m)\left[(1l)^2 - 1l \times 6m + (6m)^2\right]$

$= 2(1l+6m)\left[1l^2 - 6lm + 36m^2\right]$

⑦ $a^3 + \frac{8}{a^3} =$

$\rightarrow$ $a^3 + \left[\frac{2}{a}\right]^3 = \left(a + \frac{2}{a}\right)\left[a^2 - a \times \frac{2}{a} + \left(\frac{2}{a}\right)^2\right]$

$= \left(a + \frac{2}{a}\right)\left[a^2 - 2 + \frac{4}{a^2}\right]$

⑧ $1 + \frac{q^3}{125} = 1^3 + \left[\frac{q}{5}\right]^3$

$\rightarrow$ $= \left(1 + \frac{q}{5}\right)\left[1^2 - 1 \times \frac{q}{5} + \left[\frac{q}{5}\right]^2\right]$

$= \left(1 + \frac{q}{5}\right)\left[1 + \frac{q}{5} + \frac{q^2}{25}\right]$

Practice Set 6.3

(i) $\boxed{a^3 - b^3 = (a-b)(a^2 + ab + b^2)}$

1. Factorise.

① $y^3 - 27 = y^3 - 3^3$

$\rightarrow \quad = (y-3)\left[y^2 + y \times 3 + 3^2\right]$

$= (y-3)\left[y + 3y + 9\right]$

② $x^3 - 64y^3 = x^3 - (4y)^3$

$= (x - 4y)\left[x^2 + x \times 4y + (4y)^2\right]$

$(x - 4y)\left[x^2 + 4xy + 16y^2\right]$

③ $27m^3 - 216n^3 = (3m)^3 - (6n)^3$

$= (3m - 6n)\left[(3m)^2 + 3m \times 6n + (6n)^2\right]$

$= (3m - 6n)\left[9m^2 + 18mn + 36n^2\right]$

④ $125y^3 - 1 = (5y)^3 - 1$

$= (5y - 1)\left[(5y)^2 + 5y \times 1 + (1)^2\right]$

$= (5y - 1)\left[25y + 5y + 1\right]$

⑤ $8p^3 - \frac{27}{p^3} = (2p)^2 - \left(\frac{3}{p}\right)^3$

$= \left(2p - \frac{3}{p}\right)\left[(2p)^2 + 2p \times \frac{3}{p} + \left[\frac{3}{p}\right]^2\right]$

$= \left(2p - \frac{3}{p}\right)\left[4p^2 + 6 + \frac{9}{p}\right]$

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⑥ $343a^3 - 512b^3 = (7a)^3 - (8b)^3$
$= (7a-8b)\left[(7a)^2 + 7a\times 8b + (8b)^2\right]$
$= (7a-8b)\left[49a^2 + 56ab + 64b^2\right]$

⑦ $64x^3 - 729y^3 = (4x)^3 - (9y)^3$
$= (4x-9y)\left[(4x)^2 + 4x\times 9y + (9y)^2\right]$
$= (4x-9y)\left[16x^2 + 36xy + 81y^2\right]$

⑧ $16a^3 - \frac{128}{b^3} = 16\left(a^3 - \frac{8}{b^3}\right)$
$= 16\left(a^3 - \left(\frac{2}{b}\right)^3\right)$
$= 16\left(a + \frac{2}{b}\right)\left[a^2 + a\times\frac{2}{b} + \left(\frac{2}{b}\right)^2\right]$
$= 16\left(a + \frac{2}{b}\right)\left[a^2 + \frac{2a}{b} + \frac{4}{b^2}\right]$

2. Simplify :

(i) $(x+y)^3 - (x-y)^3$
$= x+y-(x-y)\left[(x+y)^2 + (x-y) + (x-y)^2\right]$
$= \cancel{x}+y-\cancel{x}+y\left[x^2+2xy+y^2+(x^2-y^2)+(x^2-2xy+y^2)\right]$
$= 2y\left[x^2+2xy+y^2+x^2-y^2+x^2-2xy+y^2\right]$
$= 2y\left[x^2+x^2+x^2+y^2-y^2+y^3+2xy-2xy\right]$

② $(3a+5b)^3-(3a-5b)^3$

$\rightarrow = (3xy-2ab) - (3xy+2ab)\left[(3xy-2ab)^2+(3xy+2ab)(3xy-2ab)\right.$

$\left.(3xy+2ab)^2\right]$

$= 3xy-2ab-3xy+2ab\left[(3xy)^2+2(3xy)(2ab)+(2ab)^2+\right.$

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Practice Set 6.4

1. Simplify

1) $\dfrac{m^2-n^2}{(m+n)^2} \times \dfrac{m^2+mn+n^2}{m^3-n^3}$

$\Rightarrow \quad = \dfrac{\cancel{(m+n)}\cancel{(m-n)}}{(m+n)+(m+n)} \times \dfrac{\cancel{m^2+mn+n^2}}{\cancel{(m-n)}\cancel{(m^2+mn+n^2)}}$

$= \dfrac{1}{m+n}$

2) $\dfrac{a^2+10a+21}{a^2+6a-7} \times \dfrac{a^2-1}{a+3}$

$\Rightarrow = \dfrac{(a+3)\cancel{(a+7)}}{\cancel{(a+7)}(a-1)} \times \dfrac{a^2-1}{a+3}$

$= \dfrac{\cancel{(a+3)}}{\cancel{(a-1)}} \times \dfrac{(a+1)\cancel{(a-1)}}{\cancel{(a+3)}}$

$= a+1$

3) $\dfrac{8x^3-27y^3}{4x^2-9y^2}$

$\Rightarrow = \dfrac{(2x)^3-(3y)^3}{(2x)^2-(3y)^2}$

$= \dfrac{\cancel{(2x-3y)}[(2x)^2+(2x)(3y)+(3y)^2}{(2x+3y)\cancel{(2x-3y)}}$

$= \dfrac{4x^2+6xy+9y^2}{2x+3y}$

4) $\frac{x^2-5x-24}{(x+3)(x+8)} \times \frac{x^2-64}{(x-8)^2}$

$\Rightarrow \quad = \frac{\cancel{(x+3)(x-8)}}{\cancel{(x+3)}(x-8)} \times \frac{x^2-8^2}{(x-8)(x-8)}$

$= \frac{\cancel{(x+8)(x-8)}}{\cancel{(x+8)(x-8)}}$

$= 1$

5) $\frac{3x^2-x-2}{x^2-7x+12} \div \frac{3x^2-7x-6}{x^2-4}$

$\Rightarrow \quad = \frac{3x^2-3x-2x-2}{(x+3)(x-4)} \div \frac{3x^2-9x+2n-6}{x^2-2^2}$

$= \frac{3x(x-1)+2(x-1)}{(x-3)(x-4)} \div \frac{3x(x-3)+2(x-3)}{(x+2)(x-2)}$

$= \frac{(x-1)(3x+2)}{(x-3)(x+4)} \times \frac{(x+2)(x-2)}{(x-3)(3x-2)}$

$= \frac{(x-1)(x+2)(x-2)}{(x+3)(x+4)(x+3)}$

$= \frac{(x+1)(x+2)(x+2)}{(x-4)(x+3)^2}$

6) $\frac{4x^2-11x+6}{16x^2-9}$

$\Rightarrow \quad = 4x - 8x - 3x + 6$

$= 4x(x-2) - 3(x-2)$

$= (x-2)(4x-3)$

$16x^2-9 = (4x)^2(3)^2$

$\boxed{\begin{array}{l} 4 \times 6 = 24 \\ 3 \times 8 = 24 \\ -3-8 = 11 \end{array}}$

$= \frac{(x-2)\cancel{(4x-3)}}{(4x+3)\cancel{(4x-3)}}$

$= \frac{(x+2)}{(4x+3)}$

7) $\dfrac{a^3-27}{5a^2-16a+3} \div \dfrac{a^2+3a+9}{25a^2-1}$

8) $\dfrac{1-2x+x^2}{1-x^3} \times \dfrac{1 \times x+x^2}{1+x}$

$\Rightarrow = \dfrac{x^2+2x-1}{1-x^3} \times \dfrac{1 \times x \times x^2}{1+x}$

$= \dfrac{(\cancel{1-x})(1-x)}{(\cancel{1-x})(\cancel{1+1x+x^2})} \times \dfrac{\cancel{1 \times x \times x^2}}{1+x}$

$= \dfrac{1-x}{1+x}$